# ProactiveEngine: Proactive Learning with Lifelong Memory via World Models

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## Abstract

Large language models have shown promise in assisting humans, but most real-world AI remains reactive, responding only to explicit user prompts. *Current proactive AI agents often fail* in *multimodal reasoning over long-term memory, which requires agents performing temporal reasoning over long-horizon multimodal histories and continual knowledge update*, and thus struggle to predict users' needs and intent. To address this challenge, we present **ProactiveEngine**, the first foundation model designed for proactive multimodal agents, which generates proactive responses in long-horizon settings. ProactiveEngine supports both video plus text and text-only input. Our model achieves an F1-score of 83.3%, substantially outperforming current commercial models. Our model also achieves strong performance on the public Proactive Benchmarks in a zero-shot setting without further training, while enabling a 7.8× increase in token context window and over 100× more historical events for long-horizon proactive prediction. We also introduce **NeuroAuroraBench**, a multimodal benchmark designed to evaluate long-term memory proactive prediction. The benchmark resulted in 24.8k annotated sessions capturing dynamic changes.

## 1 Introduction

The pursuit of Artificial General Intelligence represents one of the most ambitious goals in computer science (Mitchell, 2024), yet current AI remains fundamentally reactive, responding to explicit prompts rather than proactively engaging with its environment (Pezzulo et al., 2024a). Growing evidence suggests that proactive learning is a prerequisite for AGI, supported by two scientific foundations: (i) neuroscientific accounts of predictive processing and active inference, where cognition arises from continuously generating predictions and minimizing prediction error through

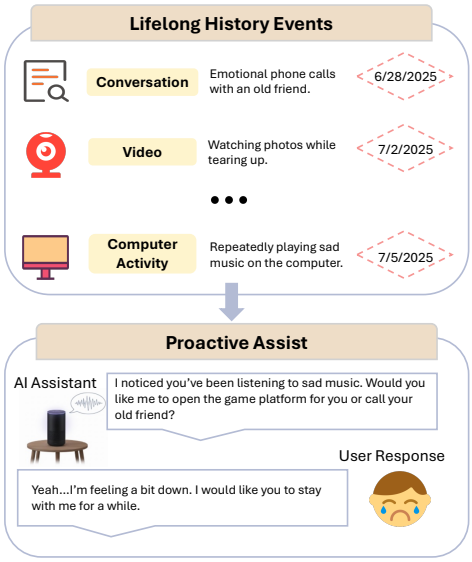



Figure 1: Proactive AI agent through users' historical events to offer warm companionship and take timely actions.

proactive engagement (Friston, 2010; Friston et al., 2017; Pezzulo et al., 2024b); and (ii) consciousness theories and empirical findings linking conscious experience to proactive predictive modeling of sensory interactions (Seth and Bayne, 2022; LeDoux et al., 2023; Grisoni et al., 2024; Matar and Marantz, 2025). Together, these insights motivate a shift from reactive systems toward proactive learning architectures capable of autonomous prediction and self-directed cognition (Pezzulo et al., 2024a).

However, these theoretical principles have not yet been fully realized in existing AI research. Despite the neuroscience-inspired promise of proactive learning, recent progress in LLM-based proactive agents has only partially captured this paradigm. In practice, current LLM-based proactive AI agents still exhibit fundamental limitations that prevent them from achieving the predictive processing capabilities essential for real proactive learning. There are 3 critical constraints: (a) Mini-

Table 1: Comparisons of NeuroAuroraBench with Existing Benchmarks.

| Benchmark | #Events | Multimodal | Exocentric | Long-term Memory | Conversation | Proactive |
|---|---|---|---|---|---|---|
| ContextAgentBench | 1000 | ✓ | ✗ | ✗ | ✓ | ✓ |
| HoloAssist | 350 | ✓ | ✗ | ✗ | ✗ | ✓ |
| ProactiveBench | 6790 | ✗ | ✗ | ✗ | ✗ | ✓ |
| LongMemEval | 500 | ✗ | ✗ | ✓ | ✓ | ✗ |
| **NeuroAuroraBench (Ours)** | **24800** | ✓ | ✓ | ✓ | ✓ | ✓ |

mal multimodal reasoning capabilities for complex environmental understanding, and (b) Lack of temporal reasoning and continual knowledge updates, preventing proactive agents from accurately predicting user needs. (c) Inability to reason over long-horizon histories. Prior proactive agents are fundamentally limited to short-term or single-turn context, and thus cannot perform predictive reasoning over long-horizon user histories. As a result, they fail to capture how user intent, habits, and needs evolve over time, making proactive predictions unstable (Yang et al., 2026; Lu et al., 2025; Yang et al., 2025a; Bandyopadhyay et al., 2025). This contradicts the predictive processing principle that intelligence emerges from minimizing long-term prediction error over latent world and user states (Friston, 2010).

There arises a key research question that remains unsolved: **Can proactive AI reason over long-term multimodal histories with temporal reasoning and continual knowledge updates, and still make correct proactive predictions?** To address these challenges and answer this research question, we introduce **ProactiveEngine**, an end-to-end foundation model for proactive AI agent that closes the loop of user need prediction, and proactive action execution with long-term memory. Unlike all prior proactive AI, which operate under short-term memory and optimize for user acceptance rate. ProactiveEngine is designed for long-term memory and multimodal prediction accuracy. In addition, it enables three new capabilities: proactive action without human prompts, proactive self-discovery that allows agents to autonomously discover real-world knowledge from evolving contexts, and proactive self-evolution, as shown in Appendix D.

To assess proactiveness, we propose **NeuroAuroraBench**, the first long-term memory benchmark for proactive AI agents and foundation models. By requiring the agent to reason over long-term memories, it increases reasoning difficulty and better reflects real-world scenarios. It contains 24.8k annotated sessions and supports 4 unique and foundational dimensions: temporal reasoning, knowledge update, multimodal reasoning, and multi-session reasoning, enabling reasoning across time and modalities. It is the only benchmark that simultaneously supports multimodal input, exocentric video, local computer activity, long-term memory, conversational history, and proactive prediction, as shown in Table 1.

Experiments on both NeuroAuroraBench and public benchmarks show that, compared with closed-source and open-source baselines, our method significantly improves the user prediction accuracy, enabling more effective proactive assistance. Our main contributions are summarized as follows:

1. We introduce *multimodal reasoning over long-term memory*, a new setting in which proactive AI agents require temporal reasoning over long-horizon multimodal interaction histories and continual knowledge updates.
2. We propose ProactiveEngine, a novel foundation model, achieving up to 83.3% prediction F1-score over the baselines. ProactiveEngine is trained as the largest foundation model to date for proactive AI in terms of context size and historical events, supporting a 7.8× larger context window and retaining on average over 100× more historical events than existing baselines.
3. Experiments further demonstrate strong generalization: models trained in our long-term memory setting transfer zero-shot to the public benchmarks with strong performance.
4. We open-source NeuroAuroraBench, the first large-scale and long-term memory benchmark for proactive AI. By combining 4 unique dimensions, 6 different help types. This benchmark contains 24.8k annotated sessions.

## 2 Related Work

**Vision-Language Model** Vision-Language models research provides fundamental insights into proactive intelligence by grounding language in

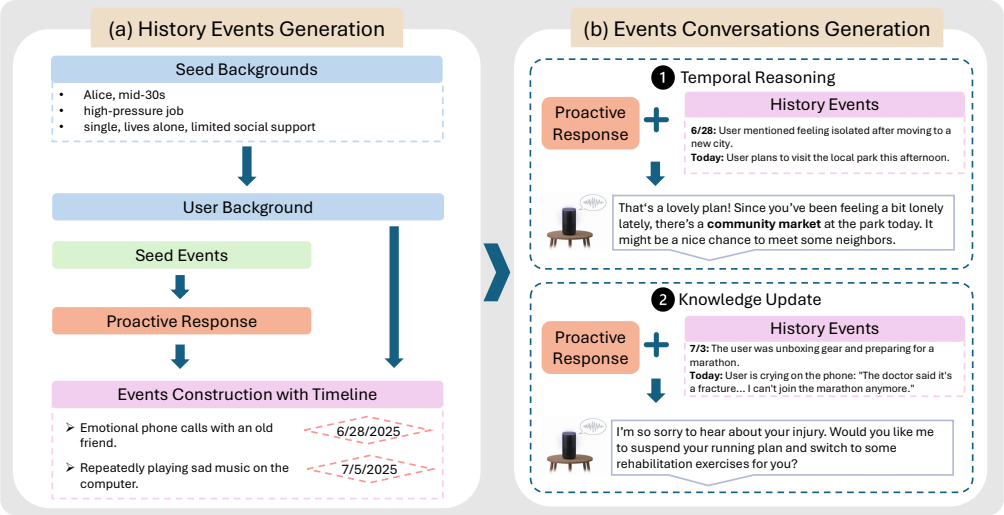



Figure 2: **Data generation pipeline:** Each step produces a single event, which can be a conversation, a video, or a computer activity on devices.

visual perception, enabling agents to anticipate user needs without prompts. Representative VLMs span unified pretraining (Alayrac et al., 2022; Li et al., 2023), stronger open backbones with long-context/video reasoning (Qwen2.5-VL, Qwen3-VL) (Bai et al., 2025b,a). VLMs have also been studied in visual reasoning settings(Song et al., 2025). In embodied settings, Colosseum-V2 further benchmarks the generalization capabilities of vision-language-action models(Morgan et al., 2026). VLMs have also been applied to text-guided video matting and model calibration(Yang et al., 2025b; Yang and Song, 2021).

**LLM-based Proactive Agents** Despite theoretical promise from neuroscience, current LLM-based proactive agents exhibit fundamental limitations. Existing systems demonstrate shallow memory access and limited temporal reasoning across interaction sessions (Yang et al., 2026; Lu et al., 2025). They lack sufficient multimodal reasoning capabilities for complex environmental understanding (Yang et al., 2025a) and cannot effectively learn from past interactions to adapt behavioral strategies (Bandyopadhyay et al., 2025). Prompt-based adaptation further improves the flexibility of large vision models (Xiao et al., 2026), while agent harness engineering extends such capabilities toward structured agentic systems (Li et al., 2026). Most critically, current agents are limited to single input modalities and short time windows, restricting their ability to determine whether users require assistance in real-world contexts.

## 3 NeuroAuroraBench Dataset Construction

**Overview** We adopt the incremental event generation strategy. At each step, the system generates one event, which can be either a history event or an evidence event.

### 3.1 Seed Backgrounds Construction

We employ a Xiaomi smart speaker Pro (Xiaomi) device to collect real-world proactive interactions in diverse daily scenarios. During each session, the device proactively initiates an assistant response based on the current context. If the user chooses to respond or confirms the intervention, we record the resulting proactive response and a synchronized 1-minute exocentric video clip. All recordings are collected from 30 volunteer participants in different environments under informed consent. For each clip, we infer a coarse activity context using a vision–language model, which is then used to condition the construction of seed user backgrounds with basic demographic and behavioral attributes. We subsequently sample and assemble a coherent user profile from this seed pool. We further divide proactive responses into 6 types: emotional support, no support, safety and health care, proactive reminders, personalization, and proactive action within the proactive self-discovery gym. Detailed distribution specifications are provided in Figure 8.

### 3.2 History Events Construction

Using collected proactive responses, user backgrounds as input, as shown in Figure 2. We construct corresponding historical event timelines that

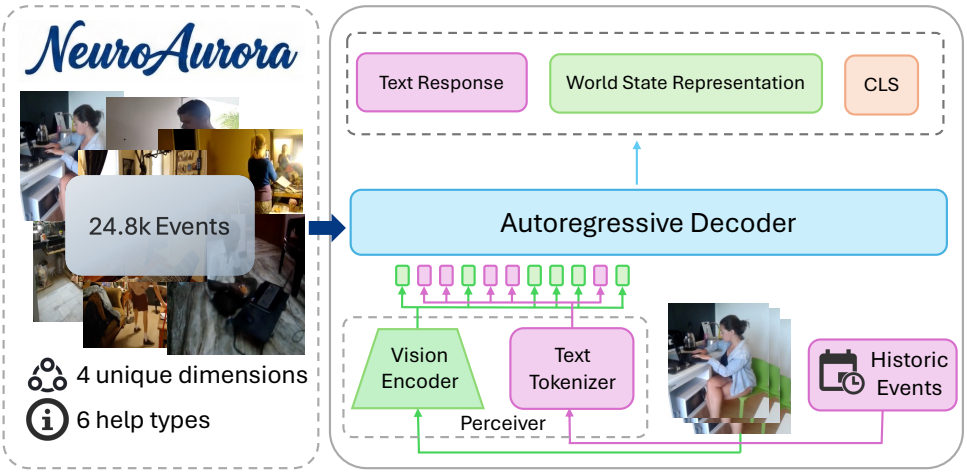



Figure 3: The framework of **ProactiveEngine**, a autoregressive transformer architecture that unifies vision and text in a shared token space to generate proactive responses, latent world state representations, and a binary proactive classification result.

necessitate these responses, building upon LongMemEval's framework(Wu et al., 2024). For each proactive response, we generate up to 2k supporting events that collectively justify the target action. Each evidence event may take the form of a visual observation, a user-AI conversation, or a local computer activity log. The events generation template could reference our Appendix C. Our generation method encourages diverse contextual signals and simulates realistic user-agent interaction settings. To model realistic environmental complexity, we integrate four essential reasoning dimensions:

**Temporal Reasoning:** The evidence supporting a proactive response is distributed across multiple events over time. Each history event incrementally introduces relevant cues, and only by reasoning over the entire sequence can the agent determine the appropriate proactive response. **Knowledge Updates:** Incorporating evolving user preferences and information changes over time. **Multimodal Reasoning** and **Multi-session Reasoning**, which focus on address cross-modal grounding and multi-session dependency.

After generating the evidence events, we further enrich the user timeline by randomly inserting normal activity events. These events serve to increase reasoning difficulty while enhancing the realism of user behavior simulation. All inserted events are drawn from real-world datasets: visual observation events are taken from Charades-Ego (Sigurdsson et al., 2018), events from LoCoMo(Maharana et al., 2024). Finally, we assign timestamps to all events in the timeline, ensuring a coherent temporal progression. For each sample session, we split the current video into a 40-second observation window as model input and a held-out 20-second future clip used to supervise future world-state prediction. In total, this process results in 24.8k annotated sessions. We randomly split the dataset into 80% for training and 20% for testing. To ensure data quality, we hired a large group of human experts to manually edit all data. Specifically, we verified that (1) the expressions were rewritten to be more natural and colloquial, (2) the evidence events were temporally and logically consistent, and (3) verified the correctness of proactive predictions. Approximately 42% of the sessions were human-edited.

## 4 Method

### 4.1 Problem Formulation.

We study the problem of proactive prediction for proactive learning: at each timestep $t$, the system must *anticipate* the next helpful assistant response from the user's ongoing multimodal interaction, *without* requiring an explicit user prompt at that moment. Concretely, the goal is to generate a response that aligns with the user's latent need implied by recent visual evidence and textual evidence. Importantly, when no intervention is needed, the correct behavior is to produce *no response*. At timestep $t$, the model receives a multimodal context consisting of a video segment: $\mathbf{V}_t = \{I_{t,1}, \ldots, I_{t,L}\}$ where $I_{t,l}$ denotes the $l$-th video frame in the segment and $L$ is the number of frames, and a textual context $\mathbf{c}_t$, which contains the dialogue history and computer activity information. These inputs are encoded into a unified representation: $\mathbf{X}_t = g([f_v(\mathbf{V}_t); f_c(\mathbf{c}_t)])$ where $f_v(\cdot)$ and $f_c(\cdot)$ are the visual and textual encoders, and function $g$

projects the concatenated features into a unified token space. The resulting $\mathbf{X}_t = \{x_{t,1}, \ldots, x_{t,N}\}$ is a sequence of $N$ tokens that summarizes the multimodal context. The output of the model is a proactive response represented as a sequence of text tokens: $\mathbf{y}_t = (y_{t,1}, \ldots, y_{t,M}), \quad y_{t,m} \in \mathcal{V}$ where $M$ is the response length (with $M = 0$ indicating no response), $y_{t,m}$ is the $m$-th token, and $\mathcal{V}$ is a fixed vocabulary. At last, proactive prediction is formulated as conditional sequence generation, with the model defining:

$$p_\theta(\mathbf{y}_t \mid \mathbf{X}_t) = \prod_{m=1}^{M} p_\theta(y_{t,m} \mid \mathbf{X}_t, y_{t,<m}) \tag{1}$$

where $\theta$ denotes the model parameters and $y_{t,<m} = (y_{t,1}, \ldots, y_{t,m-1})$ is the prefix of previously generated tokens.

### 4.2 Architecture Design

We propose **ProactiveEngine**, a foundation model for proactive learning. As illustrated in Figure 3, the **Perceiver** encodes video observations with an image encoder and textual context with a Byte-Pair Encoding (BPE) tokenizer. These tokens are then provided to a transformer based on Qwen3-VL-8B(Bai et al., 2025a) to generate proactive responses. We use the SigLIP2(Tschannen et al., 2025) as the vision encoder, and video inputs are injected into the unified token sequence via the vision delimiters <|vision_start|> and <|vision_end|>.

Our model produces **three outputs** from the same autoregressive decoding process: *(i) proactive response text*, *(ii) a future latent world state prediction*, and *(iii) a binary intervention prediction* indicating whether proactive help is needed. Concretely, the transformer first generates the proactive response text tokens. It then predicts the latent representation of the future world state, since our objective is not to reconstruct pixels but to model the semantic future state, inspired by joint-embedding predictive architectures (JEPA)(Chen et al., 2026). We leverage the same vision encoder to encode future video into latent representations, and discretize the resulting latent features into visual tokens. Additionally, we attach an MLP classification head on top of the decoder's final-layer hidden state to perform binary prediction of whether the user needs assistance.

## 5 Experiments

Learning effective proactive responses is a highly difficult problem, as the model must identify the right intervention timing, integrate multimodal context, capture long-term historical dynamics, and scale over extended contexts at the same time. Therefore, our training method follows a two-stage pipeline. In the first stage, we extend the foundation model's effective context length by training on data with progressively increasing sequence lengths. In the second stage, we perform instruction-following via interleaved training on multimodal data for capturing dynamic changes in history. All stages are trained with 64 NVIDIA A100 GPUs. We fix the global token budget to 4M tokens per batch across all stages with a learning rate of $1 \times 10^{-4}$.

### 5.1 Stage 1: Training for Extended Context Lengths

To enable proactive prediction over extremely long multimodal interaction histories, we adopt the training paradigm of LWM(Liu et al., 2025), which allows scaling to million-length sequences with only linear memory growth, effectively extending our base model well beyond its native context size. This approach theoretically supports near-infinite context lengths, and we extend the model's context window from 256K to 1M tokens by leveraging these techniques. We also use the long-context QA data construction pipeline for Stage 1.

In addition, we adopt YaRN scaling (Peng et al., 2024) to improve stability when extrapolating positional encodings to longer contexts. Concretely, under the Interleaved-MRoPE(Bai et al., 2025a) design, we set the scaling factor to 3. We then initialize from Qwen3-VL-8B as the base model and progressively extend its effective context length across three stages: 32K, 512K, and 1M. At this stage, we train on the Book3 dataset of The Pile(Gao et al., 2020) to support stable long-context learning. Detailed statistics for each stage, including the number of tokens and training duration, are reported in Table 6. Each stage is initialized from the checkpoint of the preceding stage to ensure smooth and stable context expansion.

Table 2 below shows that expanding the context window from 32K up to 1M does not degrade performance on standard benchmarks. Across all benchmarks we tested, ProactiveEngine maintains accuracy that is comparable to or slightly better

than the base Qwen3-VL-8B, demonstrating that long-context scaling does not harm short-context reasoning. Importantly, hallucination rates measured by HallusionBench remain stable across all context lengths, indicating that enlarging the context does not introduce spurious predictions. These results validate that our model can support ultra-long contexts while preserving reliability and performance on conventional tasks.

Table 2: Performance evaluation across vision-language benchmarks, comparing baseline Qwen3-VL-8B and ProactiveEngine variants (32K to 1M).

| | | ProactiveEngine | | |
|---|---|---|---|---|
| **Benchmark** | **Qwen3-VL-8B** | **32K** | **512K** | **1M** |
| MMMU(Yue et al., 2024) | 74.1 | 74.9 | 74.8 | 73.3 |
| MMBench-EN(Liu et al., 2024) | 85.3 | 84.8 | 85.2 | 85.2 |
| MMBench-CN(Liu et al., 2024) | 84.5 | 85.5 | 84.6 | 84.4 |
| DocVQA$_{test}$(Mathew et al., 2021) | 96.1 | 96.4 | 96.3 | 95.8 |
| HallusionBench(Guan et al., 2024) | 65.4 | 65.4 | 65.2 | 65.3 |

## 5.2 Stage 2: Multimodal Training with Long-Horizon World Modeling

Having equipped the model with ultra-long context capacity in Stage one, we now proceed to train the model for proactive prediction over long-horizon multimodal histories. The primary goal of this stage is to enable the model to correctly infer when assistance is needed and what assistance should be provided, especially in scenarios where user needs emerge from temporal reasoning and continual knowledge updates. To this end, we train the model to operate under two settings: (i) text-only event streams, and (ii) text interleaved with video observations. In both cases, the model is required to generate not only an appropriate proactive response, but also a latent world representation of the predicted future world state. This design forces model to predict future states regardless of whether visual inputs are available, thereby enabling more accurate proactive prediction.

We initialize from the long-context backbone obtained in Stage 1 and perform progressive training on **NeuroAuroraBench** using both text-only event streams and text–video interaction data. Across all stages, the YaRN scaling factor is kept fixed, as the backbone already supports up to 1M tokens, which enables the model to acquire long-horizon proactive reasoning while preserving short-context competence.

We train **ProactiveEngine** using a unified objective that jointly optimizes proactive response generation, future world-state prediction, and intervention decision. Given the multimodal context, the overall training loss is defined as:

$$\mathcal{L}(\theta) = \lambda_{\text{resp}}\,\mathcal{L}_{\text{resp}}(\theta) + \lambda_{\text{future}}\,\mathcal{L}_{\text{future}}(\theta) + \lambda_{\text{int}}\,\mathcal{L}_{\text{int}}(\theta) \quad (2)$$

where $\mathcal{L}_{\text{resp}}$ and $\mathcal{L}_{\text{future}}$ are autoregressive negative log-likelihood losses for proactive response generation and future latent state prediction, respectively, and $\mathcal{L}_{\text{int}}$ is a binary classification loss for intervention prediction. All $\lambda$ terms are treated as hyperparameters. In practice, we optimize this objective under ultra-long interaction histories using the following training strategies:

1. **Curriculum over Interaction Length.** We partition NeuroAuroraBench into four subsets according to the number of historical events: (i) $\mathcal{S}_1$: fewer than $5$ events, (ii) $\mathcal{S}_2$: fewer than $50$ events, (iii) $\mathcal{S}_3$: fewer than $0.8k$ events, and (iv) $\mathcal{S}_4$: fewer than $2k$ events. Training proceeds progressively from $\mathcal{S}_1 \rightarrow \mathcal{S}_2 \rightarrow \mathcal{S}_3 \rightarrow \mathcal{S}_4$, allowing the model to first learn short-range dependencies before being exposed to long-horizon temporal reasoning, which improves stability and generalization across different memory scales.

2. **Multimodal Dropout for Text–Video Data.** To ensure the model functions both *with* and *without* visual sensors, we randomly drop the video stream for $30\%$ of training samples across all subsets. In these cases, the model is required to predict the future world representation and proactive outputs using only the textual event history. This modality dropout encourages shared semantic representations that remain consistent across text-only and text-plus-video settings. More ablation experiments about the ratio can be found in the subsection B.2.

3. **Short-Context Anchoring and Packed-Sequence Training.** During training on the long-horizon subsets $\mathcal{S}_3$ and $\mathcal{S}_4$, we additionally mix $28\%$ randomly sampled, non-overlapping short interactions from $\mathcal{S}_1$ and $\mathcal{S}_2$ into each batch. This short-context anchoring mitigates catastrophic forgetting and preserves proactive prediction performance on short histories. To efficiently train with ultra-long sequences, we pack multiple interaction segments into a single sequence and apply *block-diagonal attention masking*, ensuring that each segment attends only to itself. We further apply loss reweighting such that packed training is equivalent to standard padded training.

Table 3: Evaluation results representative models and methods on NeuroAuroraBench. All baseline models exhibit limited accuracy and F1-score performance under long-horizon proactive reasoning, while ProactiveEngine achieves a clear margin across all metrics.

| Model | Recall | Precision | Accuracy | F1-Score | False Alarm | Eval Score |
|---|---|---|---|---|---|---|
| *Commercial Baselines* | | | | | | |
| GPT-4o-mini | 80.2% | 41.1% | 45.1% | 54.3% | 78.6% | 2.1/5.0 |
| GPT-4o | 42.8% | 44.6% | 52.1% | 43.7% | 36.2% | 2.2/5.0 |
| Claude-4-Opus | 88.4% | 40.9% | 42.2% | 55.9% | **90.8%** | 2.3/5.0 |
| Claude-4.5-Opus | 72.6% | 69.8% | 68.9% | 71.2% | 24.1% | 3.6/5.0 |
| GPT-5 | 69.7% | 68.8% | 69.4% | 69.2% | 16.4% | 3.1/5.0 |
| *Open-Source Models* | | | | | | |
| InternVL3-14B | 86.3% | 52.2% | 60.4% | 65.1% | 54.2% | 2.7/5.0 |
| Qwen2.5-VL-7B | **89.1%** | 49.2% | 58.3% | 63.3% | 60.1% | 2.4/5.0 |
| Qwen2.5-VL-72B | 70.8% | 63.4% | 66.6% | 66.9% | 32.6% | 3.4/5.0 |
| LLaMA-3.1-8B | 72.1% | 60.2% | 64.2% | 65.6% | 45.2% | 3.2/5.0 |
| *Existing Methods* | | | | | | |
| Proactive Agent | 9.6% | 39.4% | 53.4% | 15.3% | 11.3% | 1.2/5.0 |
| Context Agent | 6.2% | 17.1% | 47.2% | 9.1% | 69.7% | 1.4/5.0 |
| **ProactiveEngine (Ours)** | 83.8% | **83.3%** | **81.5%** | **83.3%** | 10.1% | **4.7/5.0** |

**Proactive Task Evaluation** As shown in Table 4. In the zero-shot setting without training on any additional benchmark data, out model consistently demonstrates strong generalization across benchmarks that span very different context regimes. On ProactiveBench and ContextAgentBench, our method outperforms both the Proactive Agent and the ContextAgent, indicating more accurate and reliable proactive decision-making even when facing short or medium-length history scenarios, showing that introducing proactive reasoning does not come at the cost of long-context understanding.

Table 4: Performance comparison across different Proactive AI benchmarks.

| Benchmark | Model/Method | Acc | F1 |
|---|---|---|---|
| ProactiveBench | Proactive Agent | 66.1 | 79.1 |
| | ContextAgent | 58.2 | 66.8 |
| | ProactiveEngine(Ours) | **77.5** | **87.3** |
| ContextAgentBench | Proactive Agent | 79.9 | 81.7 |
| | ContextAgent | 89.4 | 89.9 |
| | ProactiveEngine(Ours) | **91.2** | **91.6** |

The most striking results are observed on NeuroAuroBench as shown on Figure 4. We construct 5 evaluation splits by partitioning test samples according to the length of the histories (1, 20, 60, 200, and 2000 events). As the number of events increases, both Proactive Agent and ContextAgent exhibit synchronized degradation in accuracy and F1 score, with their F1 values even **dropping to near zero** once the history exceeds 200 events, indicating that they almost completely fail to detect situations where help is actually required. At the same time, their accuracies collapse to around 50%, which corresponds to **random guessing** in this binary decision setting. In contrast, ProactiveEngine remains robust across all history scales. **This outcome reveals a fundamental limitation of existing methods when interaction histories become very long, and their performance largely degrades as the event horizon grows**.

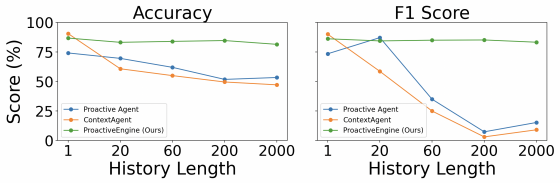



Figure 4: Context length and historic events comparison.

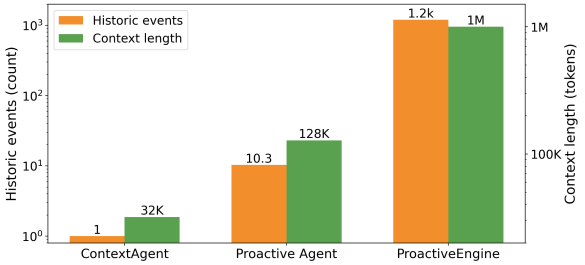



Figure 5: Context length and historic events comparison.

As illustrated in Figure 5 above, this failure reflects a fundamental limitation in how existing models utilize their representational capacity over long histories. Compared to the Proactive Agent, ProactiveEngine supports a **7.8× larger context window** (1M vs. 128K tokens) and can retain on average **over 100× more historical events**.

We further evaluate leading foundation models as shown in Table 3 on NeuroAuroraBench.

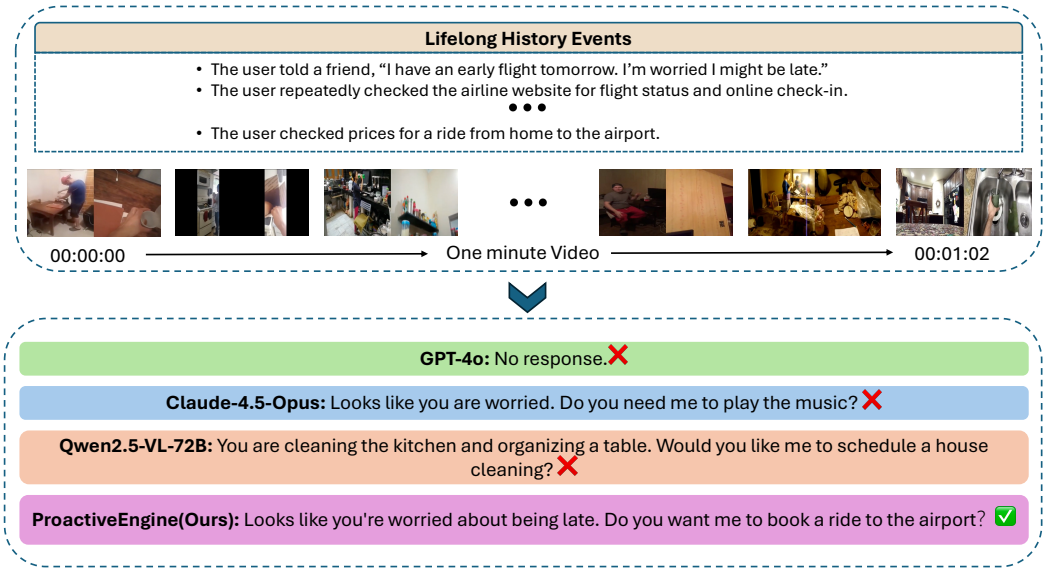



Figure 6: A representative failure case of existing models on long-horizon proactive prediction. GPT-4o fails to issue any proactive response, Claude-4.5-Opus misidentifies the user's need, and Qwen-2.5-VL-72B relies primarily on the current visual observations while ignoring long-term history, whereas ProactiveEngine successfully integrates historical events with visual observations to generate an appropriate proactive response.

Despite their strong general-purpose capabilities, these models fail to maintain reliable proactive predictions over ultra-long interaction histories, most of them exhibit performance close to random guessing under this setting. We also report standard proactive prediction metrics in this experiment, including **Accuracy**, **F1-score** and so on, following prior work (Lu et al., 2025). However, as correct proactive responses can take flexible semantic forms, exact string matching policy can lead to inaccurate evaluation. To address this issue, we further introduce an LLM-based evaluator by prompting the DeepSeek-R1(Guo et al., 2025) model to assess semantic correctness and response quality, following the G-Eval protocol (Liu et al., 2023). Our meta-evaluation shows over 97% agreement with human experts. This score is reported as the final column in our tables as Eval Score.

**Out-of-Domain Evaluation** Beyond proactive prediction, a practical foundation model must also remain reliable when users explicitly initiate queries. As a result, we also evaluate our model in out-of-domain benchmark for long-term question-answering setting, where the task focuses on temporal reasoning under continually updated knowledge histories. Unlike proactive prediction, this experiment does not require the model to decide *when* to intervene, but instead assesses whether it can produce correct responses. This capability is crucial for real-world deployment.

We evaluate our model on two long-term memory benchmarks, LongMemEval and Deep Memory Retrieval (DMR)(Rasmussen et al., 2025), with results summarized in Table 5. LongMemEval measures a model's ability to answer temporally grounded questions over extended interaction histories. DMR focuses on accurate memory retrieval across multi-session conversations. On LongMemEval, ProactiveEngine achieves a score of 62.0, outperforming strong large language model baselines, including GPT-4o and LLaMA-3.1-Instruct variants. On DMR, ProactiveEngine attains a score of 92.0, demonstrating competitive deep memory retrieval performance.

Table 5: Performance comparison on long-term memory benchmarks.

| Benchmark | Model/Method | Score |
|---|---|---|
| LongMemEval | GPT-4o | 60.0 |
| | LLaMA-3.1-Instruct-70B | 33.0 |
| | LLaMA-3.1-Instruct-8B | 45.4 |
| | ProactiveEngine (Ours) | **62.0** |
| DMR | MemGPT | 93.4 |
| | Zep | **98.2** |
| | ProactiveEngine (Ours) | 92.0 |

## 6 Conclusion

In summary, our model demonstrates that long-term memory and world modeling are key to enabling reliable proactive intelligence over long interaction histories. We hope this work will inspire future research toward truly autonomous and self-learning proactive AI.

## Limitations

ProactiveEngine does not update its parameters during deployment and relies on offline training with stored interaction context. More broadly, this reflects a limitation of our work, which remains largely constrained by pretraining and differs from human learning, where intelligence emerges through active exploration, continual adaptation, and interaction with the world. Future work could move toward constrained online learning, allowing proactive agents to safely explore open-ended environments, receive long-term feedback, and update their policies through reinforcement learning. Beyond short-term proactive prediction, such agents could be trained with broader biologically inspired objectives, such as self-maintenance, token-level resource awareness, and long-horizon goal persistence. This also represents a major research gap in current AI. This underexplored direction may provide a path toward human-level AI by treating proactiveness as a core mechanism of learning.

## Ethical Considerations

Our work studies proactive AI over long-term multimodal histories, which may include personal behavior, video, conversation, and computer-activity contexts; therefore, all data collection and release follow strict consent, anonymization, and responsible-use protocols. All real-world recordings were collected from volunteer participants under informed consent, with clear notice of the collection purpose, recorded data types, and academic research use. Any released multimodal data are privacy-preserved through face blurring/mosaicing and removal or anonymization of personally identifiable information, including real identities, private accounts, contact details, financial information, and precise location traces. Figures containing real examples are used only with participant permission. NeuroAuroraBench is released only for research on proactive reasoning and long-term memory, under terms that prohibit re-identification, sensitive-attribute inference, surveillance, covert monitoring, manipulation, and unauthorized behavioral profiling.

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## Appendix Overview

SECTION

§A **Training Details:** model size, sequence length, YaRN factor, batch tokens, total training tokens, training time, and document length buckets.

§B **Ablation Study:** controlled analysis of the world model, intervention classification head, and multimodal dropout ratio.

§C **Prompt Template:** event generation and evaluator scoring prompts used for proactive response construction and evaluation.

§D **Action Type Taxonomy:** Definitions of proactive self-discovery and proactive action, together with the API schema used in the environment.

§E **Data Distribution:** full dataset statistics and help type distribution for NeuroAuroraBench.

## A Training Details

Table 6: Training Configuration for Different Sequence Lengths

| Configuration | 32K | 512K | 1M |
|---|---|---|---|
| Model Size | 8B | 8B | 8B |
| Sequence Length | 32K | 512K | 1M |
| YaRN factor | 3 | 3 | 3 |
| Tokens per Batch | 4M | 4M | 4M |
| Total Tokens | 1.2B | 0.8B | 1.5B |
| Training Time | 240h | 120h | 72h |
| Document Length Bucket | 10K–200K | 200K–1M | 1M+ |

## B Ablation Study

### B.1 World Model and Intervention Classification Head

To isolate the contributions of our two core design choices, world model and the intervention head, we conduct controlled ablations on NeuroAuroraBench. We evaluate two variants. First, we remove future latent prediction by setting $\lambda_{\text{future}} = 0$ while keeping proactive response generation and intervention training unchanged. Second, we remove the binary intervention classifier by setting $\lambda_{\text{int}} = 0$ while keeping response generation unchanged. Following our main evaluation protocol,

we report performance under five history length splits of 1, 20, 60, 200, and 2000 events.

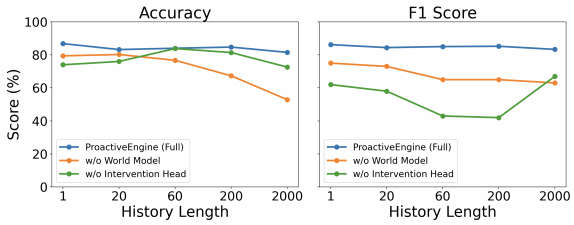



Figure 7: Context length and historic events comparison.

We observe a clear turning point as interaction histories grow long. When the history length exceeds 200 events, both models without world model or an intervention head exhibit a pronounced performance drop. This trending line clearly reveals that robust long-horizon proactive prediction benefits from modeling future latent world representations or explicitly learning when to intervene.

### B.2 Multimodal Dropout Ratio

Table 7 illustrates the trade-off introduced by the multimodal dropout ratio. When the model is trained with full video input (0:100), it achieves the strongest performance on NeuroAuroraBench, indicating that visual signals are highly informative for in-domain proactive prediction. However, this setting generalizes poorly to external benchmarks, suggesting an over-reliance on visual context. As the proportion of text-only samples increases, performance on ProactiveBench and ContextAgentBench steadily improves, reflecting stronger robustness to missing visual inputs. Notably, the 30:70 text-to-video ratio yields the most balanced results, maintaining high accuracy on NeuroAuroraBench while significantly boosting performance on both external benchmarks. We therefore adopt this ratio as the default setting in all experiments.

Table 7: Effect of multimodal dropout ratio on proactive prediction accuracy across different benchmarks.

| Text : Video | Proactive Bench | ContextAgent Bench | NeuroAurora Bench |
|---|---|---|---|
| 0 : 100 | 53.4 | 66.1 | **83.0** |
| 10 : 90 | 64.6 | 78.3 | 82.4 |
| 30 : 70 | **77.5** | 91.2 | 81.5 |
| 50 : 50 | 75.1 | **91.5** | 76.8 |
| 70 : 30 | 73.4 | 88.0 | 69.9 |
| 100 : 0 | 78.2 | 92.7 | 54.3 |

### B.3 Comparison with Retrieval-Based Baselines

To further disentangle the effect of context length from the contribution of our method, we evaluate context-compressed long-history baselines under the same 128K context budget. Specifically, we divide each long interaction history into windows of 100 events, summarize each window independently, and concatenate the resulting summaries as the compressed history input. For the longest sessions with 2,000 events, this produces 20 compressed window summaries, allowing the full history to fit within the 128K-token limit.

As shown in Table 8, context-compressed baselines still perform substantially worse than ProactiveEngine. Although summarization allows standard VLMs to access a compressed version of the full history, it often removes temporally distant but decision-critical evidence, which is essential for long-horizon proactive reasoning. As a result, both Qwen2.5-VL-72B and Claude-4.5-Opus achieve much lower F1 scores and significantly higher false-alarm rates. In contrast, ProactiveEngine maintains a high F1 score of 83.3 and a low false-alarm rate of 10.1, demonstrating that its advantage comes not merely from a larger context window, but also from its training design for long-horizon proactive reasoning.

| Model / Baseline | F1-Score | False Alarm |
|---|---|---|
| Qwen2.5-VL-72B | 61.4 | 73.2 |
| Claude-4.5-Opus | 55.6 | 48.9 |
| ProactiveEngine (Ours) | **83.3** | **10.1** |

Table 8: Comparison with context-compressed long-history baselines under the same 128K context budget. ProactiveEngine substantially outperforms compressed-history baselines, indicating that its gains are not solely due to context length.

## C Prompt Template

### C.1 Evidence Event Generation Template

The prompts in this section are used to generate each iterative event within our environmental gym as followed:

```
You are an event generator: using all
    prior_events as context, you
    emit
exactly one new, time-stamped micro-
    observation.

====
```

```
GOAL
====

For each call you receive

1. prior_events : an ordered list of
    events produced in previous turns
   (may be empty on the first call).

2. simulator_feedback : optional
    string (may be empty).

Your job is to output exactly ONE new
     event that explicitly revises,
refines, or contradicts an earlier
    fact (a true knowledge update).
    The
newest fact always supersedes older
    ones.

After emitting the event, decide
    whether the full chain now
    resolves all
outstanding fact conflicts and
    uniquely justifies the
    proactive_response;
if yes, set "complete": true,
    otherwise "complete": false.
```

Due to space limitations, this section contains a total of 4 pages of prompts, which will not be further included here.

### C.2 Evaluator Score Template

We then introduce the Evaluator Score, which is used to assess the quality of proactive responses generated by large language models. This metric compares each proactive response generated by the model with the corresponding ground truth response. A score from 1 to 5 is assigned based on predefined criteria that consider both content alignment and tone/style consistency. The evaluation is conducted using a prompt-based scoring framework. The full prompt is shown below:

```
<Role>
You are an impartial evaluator who
    scores how well a model's
    proactive
response matches the ground-truth (GT
    ) proactive response.
</Role>

<Instructions>
You will receive:
1. **events** - recent user-event
    history (optional, aids judging
    relevance).
2. **gt_response** - the reference
    proactive reply written by humans
    .
3. **candidate_response** - the model
    's proactive reply to be graded.

Your task:
■ Compare **content** (ideas, actions
     proposed, key details).
■ Compare **tone/style** (politeness,
     empathy, formality, phrasing).
■ Then assign a **single integer
    score 1-5** per the rubric below
   and give a brief justification (≤
       50 words).

<Rubric>
5 - Perfect: content fully matches GT
     and tone/style matches.
4 - Good   : content fully matches,
    but tone/style noticeably differs
    .
3 - Fair   : main idea matches but
    some details differ OR minor
    omissions.
2 - Poor   : proactive response given
     yet content clearly wrong.
1 - Irrelevant: response unrelated to
     scene and shares no meaningful
    overlap.

<Output-Format>
Return JSON only:

{
  "score": <1-5>,
  "justification": "<your brief
      reasoning>"
}

Do not output anything else.
</Instructions>

<Example-Call>
{
  "events": [ "... optional events
      ..." ],
  "gt_response": "Don't forget to
      take your 2 pm medication.",
  "candidate_response": "Remember to
      stay positive-you're doing
      great!"
}

<Expected-Output-Example>
{
  "score": 2,
  "justification": "Provides
      emotional support instead of
      medication
                    reminder; tone
                        friendly but
                        content
                        mismatched."
}
```

## D Action Type Taxonomy

To structure the behavioral capabilities of the proactive agent, we define an action taxonomy consisting of two high-level categories: **proactive self-**

**discovery** and **proactive action**.

**Proactive Self-Discovery** refers to actions in which the agent proactively explores or queries the environment without causing any changes to the world state. These behaviors are designed to help the agent build awareness of the current context. Typical examples include `google_search`, `get_datetime`, and `check_calendar_conflict` and so on. Such operations update the agent's internal understanding but do not lead to any direct side effects or alterations of the external environment.

**Proactive Action** refers to behaviors that directly and explicitly alter the external world state. These actions are designed not just to observe or retrieve information, but to effect concrete changes in the environment. For instance, executing `play_music` updates the music status variable from `None` to a specific track with associated metadata. Similarly, invoking `book_uber` attempts to call a ride, which, if successful, modifies the uber status field in the world state.

Unlike proactive self-discovery, proactive actions always carry the potential to transition the world from one world state to another. Crucially, even when the same action is issued, different underlying world states may result in different state transitions and feedback. For example, booking a ride late at night in a remote area may lead to a failed request, whereas the same action during the day in a city center may succeed. This context sensitivity is fundamental to the agent's ability to act realistically in dynamic environments.

All attributes and arguments are shown on Table 9.

Table 9: Proactive Discovery & Action API Schema

| Category | Name | Required Arguments | Optional Arguments |
|---|---|---|---|
| **proactive_discovery** | get_weather | location, time | — |
| | get_date_time | — | — |
| | check_time_conflict | — | — |
| | google_search | query | — |
| **proactive_action** | book_uber | origin, destination | car_type (standard [default] or luxury) |
| | add_meeting | start_time, end_time or start_time + duration | — |
| | play_music | — | genre, song_name, artist_name |
| | send_email | recipient, content | subject |
| | adjust_temperature | temperature_celsius | — |
| | turn_on_light | location *or* light_id | — |

## E Data Distribution

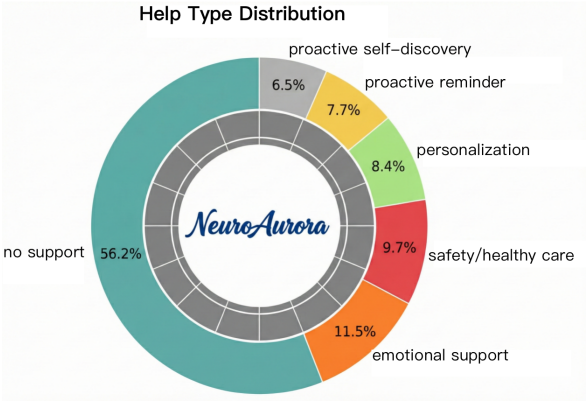



Figure 8: Help Type Distribution for NeuroAuroraBench.

Table 10: **Full dataset statistics.** Events-per-session summary for NeuroAuroraBench. The benchmark contains 24,800 sessions in total.

| Statistic | Min | Mean | Max |
|---|---|---|---|
| #Events per session | 1 | 1,217 | 2,000 |